

2016.0 RANGE ROVER (LG), 413-01

INSTRUMENT CLUSTER

DIAGNOSIS AND TESTING

PRINCIPLES OF OPERATION

For a detailed description of the instrument cluster operation, refer to the relevant description and operation section in the workshop manual. REFER to: [Instrument Cluster](#) (413-01 Instrument Cluster, Description and Operation).

CAR CONFIGURATION FILE (CCF)

CAUTION:

If a new instrument cluster is to be installed, the instrument cluster renewal procedure must be carried out using the approved diagnostic system. This will ensure that the CCF data is correctly transferred from the body control module to the replacement cluster. The CCF will also need to be updated using the approved diagnostic system if the vehicle is modified in service from its original factory specification. This can include the fitting of non-standard wheels and/or tires and optional accessory dealer install components with an electrical interface, such as park distance control

The CCF contains all relevant data about the specification and market condition of the applicable vehicle, immobilization codes and driver

personal settings. This information is retained in the body control module, the engine control module and the instrument cluster enabling each system module to detect which systems and components are installed to the vehicle. The information is continuously transferred between these three system modules to ensure that the data is constantly backed-up between the modules

INSPECTION AND VERIFICATION

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault and may also cause additional faults in the vehicle being checked and/or the donor vehicle

1. Verify the customer concern
2. Confirm which, if any, warning indicators and/or messages were displayed on the instrument cluster. For a list of messages:
REFER to: [Information and Message Center](#) (413-08 Information and Message Center, Description and Operation).
3. Visually inspect for obvious electrical faults
4. With the ignition on, check the operation of the audio output from the instrument cluster integrated speakers by operating the turn signal indicators (left and right) and verifying that audible feedback (a ticking sound) is present

Visual inspection

ELECTRICAL

- Battery
- Fuses

- Passenger and battery junction boxes
- Megafuses
- Wiring harness
- Damaged, loose or corroded connectors
- CAN circuits
- Instrument cluster
- Body control module
- Engine control module

5. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step

6. Use the approved diagnostic system or a scan tool to retrieve any DTCs before moving onto the DTC index

- Make sure that all DTCs are cleared following rectification

DTC INDEX

For a complete list of all diagnostic trouble codes that could be logged on this vehicle, please refer to section 100-00.

REFER to: [Diagnostic Trouble Code Index - DTC: Instrument Cluster](#) (100-00 General Information, Description and Operation).